

Outbound Cylinder Data Feed — Integration Specification

For JP Minda's integration engineers — pulling cylinder production data into SAP

Version: 1.0 **Date:** 06 Jul 2026 **Audience:** SAP / integration team

This document describes how your systems **pull** cylinder production data out of the tracking platform — the same mechanism (and the **same API key**) you already use for the template feed at `/sync/templates`.

1 Endpoints

Base	<code>https://digitisationapi.jpmgroun.co.in/api/v1</code>
Auth	header <code>x-api-key: <SAP sync key></code> (identical to the template feed) — or <code>Authorization: ApiKey <key></code>
Method	GET only (this feed is read-only)

Endpoint	What it returns
GET <code>/sync/cylinders</code>	State stream — cylinders whose state changed since your cursor (one snapshot per cylinder: current stage, status, latest measurements)
GET <code>/sync/stage-records</code>	Event stream — every per-stage record, append-only (e.g. <code>?stage_no=1</code> = every Pipe Cutting entry operators submit)
GET <code>/sync/cylinders/{pipeId}</code>	Deep pull — the full 21-stage trace for one cylinder

2 The polling recipe (cursor-based, exactly-once)

1. First call: GET `/sync/cylinders?limit=200` (no cursor → starts from the beginning).
2. From each response take `nextCursor` and, while `hasMore` is `true`, call again immediately with `?cursor=<nextCursor>`.
3. When `hasMore` is `false`, **persist the last non-null `nextCursor`** and reuse it on your next scheduled poll (e.g. every minute). You will only ever receive what changed since.
4. Treat `cursor` as **opaque** — store it and echo it back, don't parse it.

`nextCursor` never moves backward: on a quiet poll (0 new rows) it simply echoes your position, so it is always safe to persist. It is `null` only on the very first sync of an empty system — keep polling without a cursor until data appears.

Freshness: rows enter the feed ~3 seconds after they are written (a small stability window that guarantees no record is ever skipped during concurrent writes).

Same recipe applies to `/sync/stage-records`. Its rows are immutable events; deduplicate by `recordId` if you ever re-pull.

You can also start from a point in time instead of a cursor: `?since=2026-07-06T00:00:00Z`.

3 Response shapes

GET /sync/cylinders?limit=2

```
{
  "success": true,
  "count": 2,
  "hasMore": true,
  "nextCursor": "2026-07-06T10:12:44.000Z|a1b2c3d4-...",
  "serverTime": "2026-07-06T12:00:01.123Z",
  "data": [
    {
      "pipeId": "PC-L1-260706-0001",
      "status": "in_process",
      "currentStageNo": 6,
      "currentStageName": "Heat Treatment",
      "currentMachineId": "HT-401",
      "latestData": { "heatNo": "H-2214", "od": 356, "furnace_temp": 892 },
      "startedAt": "2026-07-06T08:01:00.000Z",
      "completedAt": null,
      "updatedAt": "2026-07-06T10:12:44.000Z"
    }
  ]
}
```

GET /sync/stage-records?stage_no=1&limit=1 — each operator Pipe Cutting entry:

```
{
  "success": true,
  "count": 1,
  "hasMore": false,
  "nextCursor": "2026-07-06T08:01:00.000Z|e5f6...",
  "data": [
    {
      "recordId": "e5f6a7b8-...",
      "pipeId": "PC-L1-260706-0001",
      "stageNo": 1,
      "stageName": "Pipe Cutting",
      "machineId": null,
      "status": "ok",
      "metrics": { "heatNo": "H-2214", "batchNo": "B-102", "grade": "34CrMo4",
        "od": 356, "wall": 7.2, "cutLength": 940 },
      "source": "manual",
      "enteredBy": "<operator user id>",
      "recordedAt": "2026-07-06T08:01:00.000Z"
    }
  ]
}
```

4 Query parameters

Param	Applies to	Meaning
cursor	both streams	opaque position from the previous response's nextCursor
since	both streams	ISO-8601 timestamp alternative to cursor (first sync / recovery)
limit	both streams	page size, default 100, max 500
status	/sync/cylinders	filter: in_process completed scrapped
stage_no	/sync/stage-records	filter to one stage (1–21); 1 = Pipe Cutting entries
pipe_id	/sync/stage-records	filter to one cylinder

Errors: 401 bad/missing key • 400 unparseable cursor / since or bad stage_no • 404 unknown pipeId on the deep pull.

5 curl quick test

```
curl "https://digitisationapi.jpmpgroup.co.in/api/v1/sync/cylinders?limit=5" \
-H "x-api-key: <SAP_SYNC_KEY>"

curl "https://digitisationapi.jpmpgroup.co.in/api/v1/sync/stage-records?stage_no=1&limit=5" \
-H "x-api-key: <SAP_SYNC_KEY>"
```